A frozen 3D U-Net is adapted to brain-MRI tasks from 32 labeled volumes each, with one LoRA module and output head per task. Whether small adapters suffice depends on where the task needs capacity, and the two task families answer oppositely - each replicated on a task chosen to share little with the original. On three segmentation tasks that inherit a large decoder but differ in channel count, pathology and scale (BraTS glioma, ISLES stroke, MSD hippocampus), full-decoder linear probing leads LoRA by a stable 0.011-0.015 Dice while training 90x more parameters. On both tasks with only a 1K-parameter head over pooled features (IXI brain age and sex, whole-head input), LoRA is the only frozen method that learns them: MAE 10.2 yr against a 14.5 yr constant baseline, and AUC 0.88, each beating full fine-tuning of 14M parameters with 56K trainable. Linear probing and BitFit emit a constant output on both, despite their frozen features ranking sex far above chance - the signal is present but a small head cannot turn it into a decision. All results use a subject-grouped held-out protocol with whole-volume inference over three seeds.